

Exercise: Population Dynamics in a Lake

Consider a lake containing three species:

- Algae
- Small Fish
- Large Fish

Let x_n represent the populations of algae, small fish, and large fish in year n .

Assume that the population evolves according to:

$$x_{n+1} = Mx_n$$

where

$$M = \begin{pmatrix} 0.80 & 0.15 & 0.05 \\ 0.15 & 0.70 & 0.15 \\ 0.05 & 0.15 & 0.80 \end{pmatrix}$$

Part 1 – Computing Future Populations

Suppose that in Year 1 the populations are

$$x_1 = \begin{pmatrix} 10 \\ 20 \\ 4 \end{pmatrix}$$

1. Compute the population vector x_2 .
2. Compute the population vector x_3 .
3. Verify that the total population remains constant.

Part 2 – Eigenvalues and Eigenvectors

1. Calculate the eigenvalues of M .
2. Calculate the corresponding eigenvectors.
3. Verify that one eigenvalue is equal to 1.
4. Show that the vector $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ is an eigenvector of M .

Part 3 – Long-Term Behavior

Compute

$$x_{30} = M^{29}x_1$$

Compare x_{30} with the eigenvector corresponding to the eigenvalue 1.

What do you observe?

Part 4 – Different Initial Conditions

Choose at least three different initial population vectors, for example:

$$\begin{pmatrix} 30 \\ 5 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} 2 \\ 50 \\ 10 \end{pmatrix}, \quad \begin{pmatrix} 100 \\ 0 \\ 0 \end{pmatrix}$$

For each initial condition:

1. Compute x_{30} .
2. Compare the resulting population distributions.
3. Explain why the long-term populations appear similar even though the initial populations were very different.